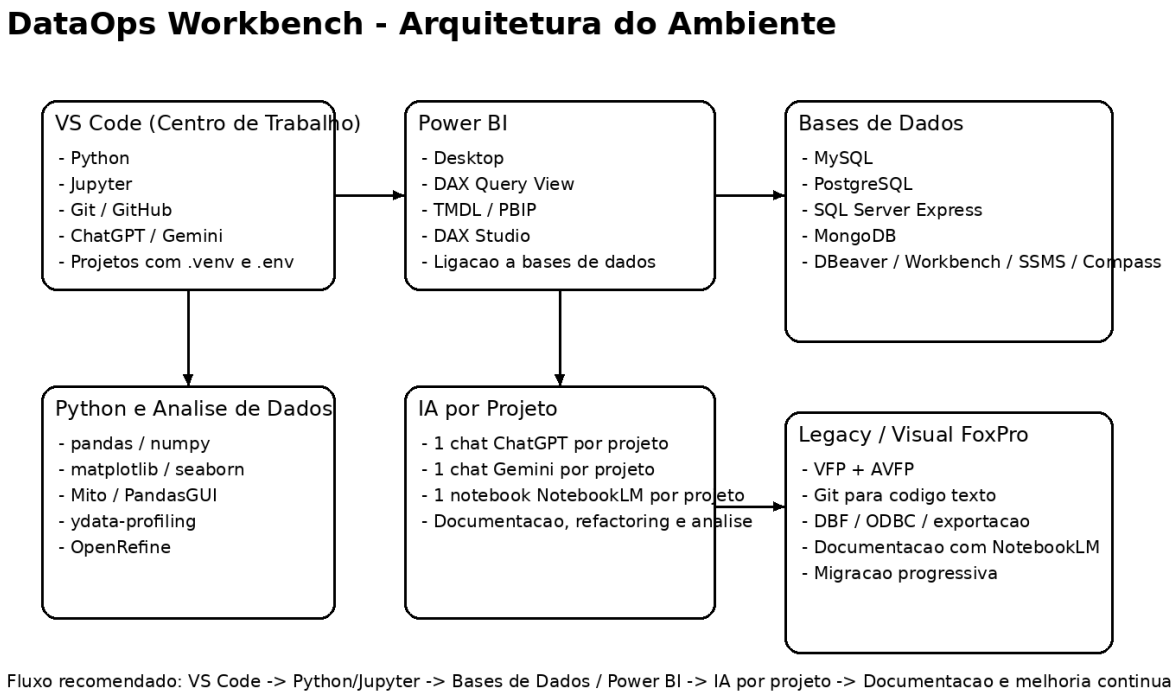
DataOps Workbench

Complete handbook for a development environment: Python, VS Code, Power BI, databases, AI and Visual FoxPro



Working edition - March 2026

Language: English

Scope: installation, configuration, usage and project-level operating model

1. Purpose and scope

This handbook defines a practical Windows-based environment for data analysis, troubleshooting, automation, reporting and gradual legacy modernization. VS Code is the center. Each project is isolated and repeatable.

1.1 What gets installed

- Python, pip, .venv virtual environments and project-level .env files
- VS Code, Python/Jupyter extensions, Codex/ChatGPT and Gemini Code Assist
- Jupyter Notebook / JupyterLab
- Git, GitHub and a standard project structure
- MySQL, PostgreSQL, SQL Server Express and MongoDB
- DBeaver as the main database client, plus Workbench, pgAdmin, SSMS, Azure Data Studio and Compass
- Power BI Desktop, DAX Studio and PBIP/TMDL in preview
- Mito, PandasGUI, ydata-profiling and OpenRefine
- NotebookLM as a project knowledge base
- Visual FoxPro support, documentation and migration guidance

2. Recommended installation order

Phase	Layer	Goal
1	Base	Python, VS Code, Python/Jupyter extensions, Git
2	Project shell	Folder structure, .venv, .env, requirements.txt, README
3	Databases	MySQL, PostgreSQL, SQL Server Express, MongoDB, DBeaver
4	Analysis tools	Mito, PandasGUI, ydata-profiling, OpenRefine
5	Power BI	Desktop, DAX Studio, Python scripting, PBIP/TMDL
6	AI	Codex/ChatGPT, Gemini Code Assist, NotebookLM
7	Legacy	Visual FoxPro, ODBC/OLE DB/DBF, AVFP, migration planning

3. Python and VS Code

Install Python from python.org and validate with python --version, py --version and pip --version. Install VS Code and the Python and Jupyter extensions first.

```
python --version
py --version
pip --version
```

3.1 Create a project

```
C:\Projects\my_project\
├── data\
├── notebooks\
├── scripts\
├── outputs\
├── pbix\
├── docs\
├── .venv\
├── .env
├── requirements.txt
└── README.md

python -m venv .venv
.\.venv\Scripts\Activate.ps1
pip install pandas numpy matplotlib seaborn ipykernel python-dotenv
```

3.2 Switching projects

- Open the correct folder in VS Code.
- Select the correct Python interpreter.
- Activate the correct .venv in the terminal.
- Make sure the .env and requirements.txt belong to the current project.

4. Git and GitHub for beginners

Git tracks changes locally. GitHub stores repositories online and supports collaboration.

```
git config --global user.name "Your Name"
git config --global user.email "you@example.com"
git init
git add .
git commit -m "Initial commit"

git remote add origin REPOSITORY_URL
git branch -M main
git push -u origin main
```

5. Databases

Engine	Recommended use
MySQL	Easy starting point, broad support, good fit for small and medium local work
PostgreSQL	Stronger relational engine, great for serious SQL

	learning and analytics
SQL Server Express	Excellent fit for the Microsoft and Power BI ecosystem
MongoDB	Document database for logs, JSON documents and semi-structured data

5.1 MySQL

Prefer MySQL 8.4 LTS for stability. Keep in mind that MySQL Installer is tied to the 8.0 series, while newer server lines also provide their own MSI packages.

```
SELECT VERSION();
SHOW DATABASES;

CREATE DATABASE demo;
CREATE USER 'app_user'@'localhost' IDENTIFIED BY 'PasswordForte123!';
GRANT ALL PRIVILEGES ON demo.* TO 'app_user'@'localhost';
```

5.2 PostgreSQL

```
SELECT version();
```

5.3 SQL Server Express

```
SELECT @@VERSION;
```

5.4 MongoDB

```
from pymongo import MongoClient
client = MongoClient('mongodb://localhost:27017/')
db = client['demo']
```

5.5 Database tools

Tool	Role
DBeaver	Main multi-database client for relational engines
MySQL Workbench	MySQL-specific administration
pgAdmin	PostgreSQL-specific administration
SSMS	Main SQL Server administration tool
Azure Data Studio	Lighter SQL Server tool
Compass	Main MongoDB GUI

6. Python integration

```
pip install sqlalchemy pymysql psycopg2-binary pyodbc pymongo pyadomd

from sqlalchemy import create_engine
engine = create_engine('mysql+pymysql://app_user:Password@localhost:3306/demo')
```

7. Power BI

Install Power BI Desktop, import a simple dataset, create a visual and create one DAX measure. Use DAX Query View first for testing DAX. Use DAX Studio for deeper analysis.

7.1 PBIP and TMDL

Saving as a Power BI Project (.pbip) is still a preview feature and must be enabled in Power BI Desktop options before use.

7.2 Python to Power BI

```
from pyadomd import Pyadomd
# connect to localhost:PORT exposed by Power BI Desktop
```

8. AI tools by project

8.1 Codex / ChatGPT in VS Code

Install the official Codex extension, sign in with your ChatGPT plan and use it inside the correct project folder.

8.2 Gemini Code Assist

Install Gemini Code Assist from the VS Code marketplace and sign in with your Google account.

8.3 NotebookLM

NotebookLM is not a native VS Code extension. Use it in the browser as a parallel project notebook. Add source documents, notes, PDFs, links and exported code.

8.4 Practical rule

- One main ChatGPT conversation per project.
- One main Gemini conversation per project.
- One NotebookLM notebook per project.

9. Data exploration tools

Mito is useful for Excel-like exploration inside Jupyter. ydata-profiling is useful for profiling and quick diagnostics. OpenRefine is useful for heavy manual cleaning.

10. Visual FoxPro and AVFP

Visual FoxPro 9.0 is out of support. Treat any VFP work as legacy maintenance and transition work, not as a modern strategic platform.

- Use VS Code and Git for text-based source files.
- Use DBF / ODBC / OLE DB to extract data.
- Use AI tools to document code, business rules and dependencies.

- Treat AVFP / VFP Advanced as community tooling and validate compatibility per application.

11. Maturity checklist

11.1 Basic

- Python, VS Code and Jupyter work.
- You can create a project with .venv.
- Git and GitHub are working.
- At least one database is reachable.
- Power BI imported data and created a simple measure.

11.2 Intermediate

- Every project has its own .venv, .env and README.
- Python connects to relational databases.
- Power BI reads from local databases.
- You use DAX Query View and DAX Studio.

11.3 Advanced

- Reusable scripts and templates exist.
- PBIP/TMDL with Git is used where appropriate.
- AI is used for review, documentation and refactoring.
- Legacy migration has a phased plan.

12. Official links

Python on Windows - https://docs.python.org/3/using/windows.html

VS Code Jupyter notebooks - https://code.visualstudio.com/docs/datascience/jupyter-notebooks

Power BI Desktop projects (PBIP) overview - https://learn.microsoft.com/en-us/power-bi/developer/projects/projects-overview

Power BI semantic model folder - https://learn.microsoft.com/pt-pt/power-bi/developer/projects/projects-dataset

OpenAI Codex IDE extension - https://developers.openai.com/codex/ide/

OpenAI Codex on Windows - https://developers.openai.com/codex/windows/

Gemini Code Assist setup - https://developers.google.com/gemini-code-assist/docs/set-up-gemini

NotebookLM Help - https://support.google.com/notebooklm/

MySQL Installer note - https://dev.mysql.com/downloads/installer/

MySQL Community Server 8.4 LTS - https://dev.mysql.com/downloads/mysql/8.4.html

PostgreSQL downloads - https://www.postgresql.org/download/windows/

SQL Server 2022 Express - https://www.microsoft.com/en-us/download/details.aspx?id=104781

MongoDB Compass - https://www.mongodb.com/try/download/compass

Visual FoxPro lifecycle - https://learn.microsoft.com/en-us/lifecycle/products/microsoft-visual-foxpro-90

Microsoft ODBC and Visual FoxPro driver docs - https://learn.microsoft.com/en-us/sql/odbc/microsoft-open-database-connectivity-odbc

DBeaver - https://dbeaver.io/

OpenRefine - https://openrefine.org/download

Mito install docs - https://docs.trymito.io/getting-started/installing-mito

ydata-profiling install docs - https://docs.profiling.ydata.ai/latest/getting-started/installation/